

Showing Work with a Calculator: Because you are using a calculator, it is possible to accidentally punch the wrong button. To ensure you earn partial credit, show the operation(s) that you entered into the calculator and the result.

Simple Interest Formula	Percent Error Formula
$I = prt$ where p = principal, r = rate, and t = time	$\frac{ \text{Estimate} - \text{Actual} }{ \text{Actual} } \times 100\%$

- Shawn found a deal for a kayak trip to see the bioluminescent jellyfish in the Indian River Lagoon. The deal said it was 40% off of the regular price. Shawn paid \$267 for 5 adults.

Determine the regular cost of the kayak trip for 1 adult.

work	
Regular Cost of One Adult	\$89.00

- Shanice makes an investment of \$6,850 into an account that earns 3.2% simple interest per year.

Determine the total amount that Shanice will have in the account after 8 years.

work	
Total Amount	\$8,603.60

3. A survey asked 100 middle school students the question, "How familiar and interested are you in the idea of sustainable foods?" The results of the survey are shown in the table.

Answer	Number of Students	Answer	Number of Students
I already know about sustainable foods, and it influences my food and beverage choices.	12	I have heard about it, but I don't really want to know more.	15
I have heard about sustainable foods and want to know more.	31	I have never thought of it before, and I don't really want to know more.	17
I have never thought of it before but want to know more.	25		

- a. What percentage of a circle graph will contain the response, "I have heard about it, but I don't really want to know more".

work	
Percentage	15%

- b. Find the corresponding angle measure of a circle graph for the response "I have heard about sustainable foods and want to know more." Round to the nearest whole number.

work	
Angle Measure (degrees)	112°

- c. Find the corresponding angle measure of a circle graph for the response "I have never thought of it before but want to know more." Round to the nearest whole number.

work	
Angle Measure (degrees)	90°

4. A survey asked 2,450 students which types of activities they are likely to participate in. The results of the survey are shown.

Category	Number of Students
Athletics and team sports	1,323
Art, music, or dance	882
Academics and learning	245

Create a circle diagram using the data. Show your work for partial credit.

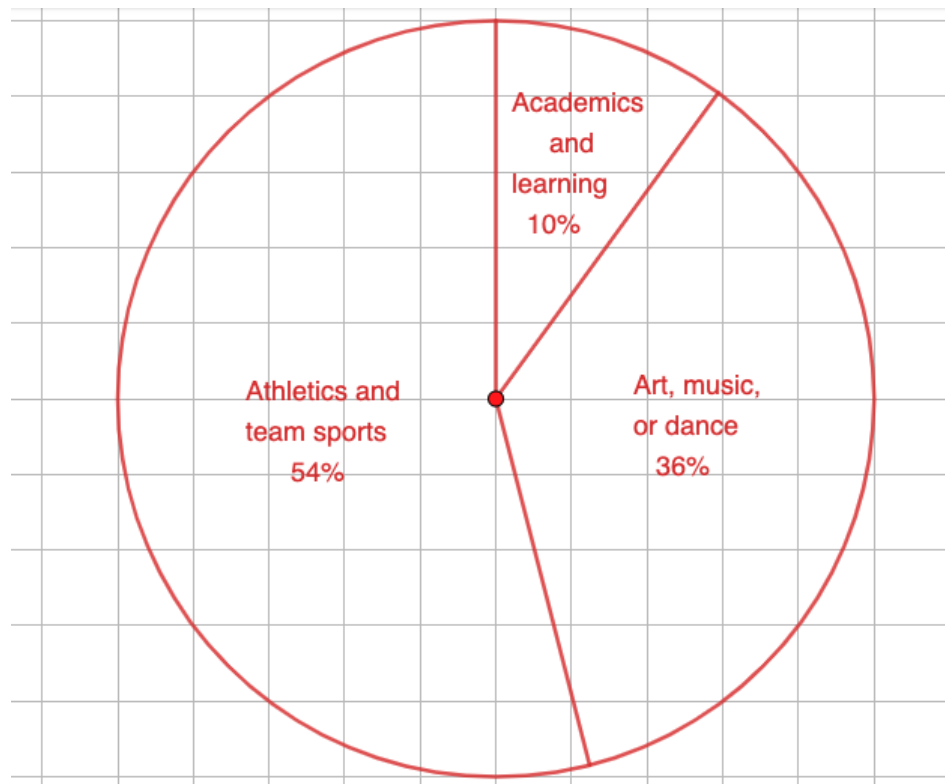
work

$$\frac{1323}{2450} = 54\% \quad 0.54 \times 360 = 194.4^\circ$$

$$\frac{882}{2450} = 36\% \quad 0.36 \times 360 = 129.6^\circ$$

$$\frac{245}{2450} = 10\% \quad 0.1 \times 360 = 36^\circ$$

Students' Preferred Activities
(title)



5. The accepted length of a tool is 282 millimeters (mm). The percent error that is allowed in the length of a tool when it is manufactured is 1.05%. During an inspection an engineer measured three tools to ensure they fit within the error. Her measurements are shown.

Tool	Measured Length	work
A	279.2 mm	$\frac{282 - 279.2}{279.2} \approx 0.010028 \dots$
B	285 mm	$\frac{282 - 285}{285} \approx 0.010526 \dots$
C	283.06 mm	$\frac{282 - 283.06}{283.06} \approx -0.003744 \dots$

Complete the statement.

Tool(s)
☐ A only
☐ B only
☐ C only
☒ A and B
☒ **A and C**
☐ B and C
☐ A, B, and C

☒ **do(es)**
☐ do(es) not

 fit within the accepted percent error.

Students could also correctly complete the statement with, "Tool(s) B only do(es) not..."

6. During his basketball career, Karl Malone took 22,975 free throws. He made 42.6% of his free throws. Determine the number of free throws Karl Malone made during his career.

work	
Free Throws Made	9,787